1.0. Overview

a) THIS IS A REQUEST FOR INFORMATION ONLY. This publication constitutes a Request for Information (RFI) as defined in Federal Acquisition Regulation (FAR) 15.201(e), "RFIs may be used when the Government does not presently intend to award a contract, but wants to obtain price, delivery, other market information, or capabilities for planning purposes. Responses to these notices are not offers and cannot be accepted by the Government to form a binding contract." This request for information does not commit the Government to contract for any supply or service whatsoever. Furthermore, the Program Office is not seeking proposals and will not accept unsolicited proposals. Responders are advised that the U.S. Government (USG) will not pay for any information or administrative costs incurred in response to this RFI; all costs associated with responding to this RFI will be solely at the interested party's expense. Not responding to this RFI may preclude participation in any further solicitation if any is issued. If a solicitation is released, it will be synopsized on the "SAM.GOV" website. (https://sam.gov/content/home). It is the responsibility of the potential respondents to monitor these sites for additional information pertaining to this requirement.

b) Information resulting from this RFI will aid Program Office personnel in determining the most appropriate acquisition strategy to meet the needs of the organization. All viable alternatives are being considered.

c) This notice is part of Government market research. Information received because of this request will be considered as Sensitive and will be protected as such. Any company or industry proprietary information contained in responses should be clearly marked as such, by paragraph, such that publicly releasable and proprietary information are clearly distinguished. Any proprietary information received in response to this request will be properly protected from unauthorized disclosure. The Government will not use proprietary information submitted from any one source to establish the capability and requirements for any future acquisition, to inadvertently restrict competition.

d) The contemplated North American Industry Classification System (NAICS) code is 336414. Responders are encouraged to suggest an alternative NAICS code if your company considers this code not the appropriate NAICS code based on the information associated with this RFI.

e) Submission of a narrative is voluntary. Respondents are advised that the United States Space Force is under no obligation to provide feedback with respect to any information submitted under this RFI.

f) The United States Space Force may or may not choose to further meet with/correspond with respondents to get further clarification of a capability.
g) Companies are advised that employees of Aerospace Corporation, a Federally Funded Research and Development Center (FFRDC) and contractors serving in the role of Systems Engineering and Technical Assistance (SETA) may review information submitted in response to this RFI for the purpose of providing technical advice to the Program Office. These employees are subject to DFARS 252.227-7025 and other applicable contract requirements, and as such, they may not further disclose proprietary information or use it for any purpose other than supporting the Program Office. Support contractors are also required to sign Non-Disclosure Agreements prior to accessing contractor data.

2.0. Description

The United States Space Force (USSF), Space Systems Command, Servicing, Mobility, and Logistics (SSC/SYD80) Program Office, is seeking information on technical concepts and approaches to Refueling Services for prepared clients in orbit.

3.0. Background and Need

As an enabling technology for the Sustained Space Maneuver need of U.S. Space Command, on-orbit servicing, including refueling, is a vital capability to support warfighter needs. Space System Command's System Engineering Review Board (SERB) has already approved two standard interfaces for satellite refueling – Northrop Grumman's Passive Refueling Module (PRM) and Orbit Fab's Rapidly Attachable Fluid Transfer Interface (RAFTI). Industry solutions for refueling of National Security Space assets equipped with these SERB-approved interfaces are sought to meet sustained space maneuver (SSM) needs by 2030.

SSC/SYD80 requests feedback from industry intending to provide services.

4.0. Questions

Please answer the following questions for your solution(s) to US SPACECOM's SSM need, enabled by on-orbit refueling:

1. Describe your solution(s) for refueling National Security Space (NSS) assets in or near Geostationary Orbits (GEO) that are at or beyond a Preliminary Design Review (PDR) or equivalent level of maturity. For each solution, please provide the following:
    a. The current maturity level, evaluated against TRL definitions (Technology Readiness Assessment Guidebook, Feb. 2025) and Technical Reviews/Audits (SMC-S-021).

        i. Design and CONOPS similarities to any prior flight-proven demonstrations, especially in GEO.

b. Provide a top-level cost and schedule estimate for advancing the Space Vehicle (SV(s)) to first flight, if applicable, and cost and schedule estimates for recurring production of subsequent units, including tailoring of cost to varying procurement approaches (e.g., schedule, number of units)
c. Identify the ability of the SV(s) to dock with and transfer propellant through SERB-approved standard refueling interfaces bi-directionally
d. Provide top-level performance parameters of the SV(s), including (but not limited to)
    i. Expected design lifespan in GEO for the SV
    ii. SV characteristics (mass, launch vehicle (LV) interface, volume envelope, hosted payload, free flying)
    iii. Total delta-v of the SV independent of transferable propellant
    iv. Compatible propellant type(s)
    v. Total amount of available transferable propellant should be at a minimum of 1000kg or more with one SV, multiple SV's may be used to provide a similar solution, or SV(s) in conjunction with an on-orbit depot
    vi. Operational propellant transfer limits (minimum and maximum)
    vii. Total number of docking & propellant transfer operations (either direction) that the SV(s) can perform
    viii. Identify the ground station and/or ground system that will be used for Telemetry, Tracking, and Commanding (TT&C) of your SV(s)
    ix. The SV and ground system must be capable of supporting an annual tempo of at least three refueling missions
    x. Your proposed solution for the SV(s), and the planned operational model (for example, Contractor-Owned, Contractor-Operated or Government-Owned, Contractor-Operated)
    xi. Range of compatible client vehicle mass & volumes with varying levels of docking & undocking probability of success
    xii. Capability of the SV to perform exquisite RPOD (ability to perform close approach through docking), propellant transfer, detach, and departure through autonomous and/or ground-controlled methods, and extent to which the SV operates autonomously for these operations
    xiii. Estimated duration in time for the RPOD operations, including detach and departure, and the estimated duration of the fuel transfer event for varying propellant transfer mass values

b. The SV is expected to adhere to industry standards and best practices for on-orbit servicing; responses should include information confirming that the SV(s) can meet client vehicle requirements such as those following:
    i. SV performs active RPOD (Rendezvous, Proximity Operations, & Docking) functions to a cooperative, passive client
    ii. Ability to manufacture and qualify SV(s) in accordance with the National Security Space Launch Port Launch Users Guide (NSSL PLUG) (NSSL-S-008-2025)
    iii. Describe how your system is/will be compliant with CNSSP No. 12
    iv. SV will ensure Do No Harm principles are applied to the entire mission design & CONOPS, including (but not limited to) collision avoidance, contamination, electrostatic discharge (ESD), electromagnetic interference & compatibility (EMI/EMC), and Orbital Debris Mitigation Standard Practices (ODMSP) compliance
    v. SV shall provide configurable CONOPS to meet client-specific needs during RPOD, mated operations, and departure, including (but not limited to) sun-pointing; active control of position and orientation; maintenance of pointing accuracy for ground contact for uplink and downlink; and command-able states for safety during the docking, refueling, and undocking phases of the mission

c. Identify your solution for a ground system and ground architecture to support operations and maintenance of SV

d. Provide nominal CONOPS as it relates to the day-to-day operations and execution of the refueling missions

2. For all solutions referenced in the above questions, please reference any upcoming plans to demonstrate the technical maturity, commercial viability, and/or cost-benefit trades for these systems on-orbit within the next 3 years

5.0. Summary

THIS IS A REQUEST FOR INFORMATION (RFI) ONLY to aid in acquisition planning, analyze the state of the market, technological and programmatic capability, business case development, budgeting, acquisition strategy and contract approach to obtain this capability on behalf of the government. The information provided in the RFI is subject to change and is not binding to the Government. All attempts will be made to ensure the most up-to-date changes are included in the RFI prior to posting. The USSF has not made a commitment to procure any of the items discussed and release of this RFI should not be construed as such a commitment or as authorization to incur cost for which reimbursement would be required or sought. All submissions are deemed government property per DoD Federal Acquisition Regulation (DFAR) and will not be returned. **NOTE: Proprietary or Confidential information will be treated as Proprietary Information (PROPIN) and marked accordingly and protected under DoD 5200.48 (Controlled Unclassified Information (CUI)).**

6.0. Submission and Narratives

6.1. Interested parties are requested to respond to this RFI with a complete response limited to 10 pages.

6.2. All responses shall be in the form of a narrative. All narratives shall state they are submitted in response to this announcement. Responses shall be submitted via e-mail to the Primary Point of Contact listed below:

Primary Unclassified Point of Contact (POC):

Wade Pilcher
Contracting Officer
Assured Access to Space (SSC/AAK)
Phone Office: (310) 292-9956
Email: wade.pilcher@us.af.mil

6.3. Narrative Due Time/Date: **Responses to this RFI are due no later than 1600 Mountain Time on 13 March 2026.**

6.4. PROPRIETARY INFORMATION (PROPIN)

6.4.1. Proprietary Information if any is provided, needs to be clearly marked as proprietary, e.g., "Company X Proprietary Information, Reason (Intellectual Property, Competition Sensitive, etc.)" and will be protected in accordance with DoD Instruction 5200.48. Responses sent will not be returned to the provider

6.4.2. To review responses, proprietary information may be given to Subject Matter Experts (SMEs) that work for the Aerospace Corporation, a Federally Funded Research and Development Center (FFRDC) and contractors serving in the role of Systems Engineering and Technical Assistance (SETA). Disclosure of PROPIN outside of government channels and their contract is STRICTLY PROHIBITED for both categories (SETA) and FFRDC support to the government that they have access to.

6.4.3. The first page of the response shall provide the following administrative information:

6.4.3.1. Industry Name, mailing address, POC, phone number, and e-mail address of a point of contact.

6.4.3.2. Respondents shall submit an electronic copy of their response in PDF format to the Primary Point of contact. Hard copy responses will not be accepted. Page size shall be 8.5 x 11 inches with Times New Roman font with 12-point font size. No marketing or company literature will be accepted as an acceptable response to this RFI.